# KELLY DIANGA

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## PROFESSIONAL SUMMARY

MSc Software Engineering student and BEng Robotics & Mechatronics graduate with hands-on expertise in embedded firmware development, hardware-software integration, and real-time systems. Proven experience in bare-metal C programming on ARM Cortex-M microcontrollers, interrupt-driven architectures, and sensor interfacing. Currently extending low-level expertise through specialised STM32 bare-metal programming training. Seeking an Embedded Systems / IoT role where I can contribute to firmware development, hardware bring-up, and embedded software engineering while gaining industry experience in a professional engineering environment.

## TECHNICAL SKILLS

**Languages:** C, C++, Python, Java, VHDL

**Microcontrollers & Platforms:** ARM Cortex-M (MK20D), ESP32, Arduino, Raspberry Pi 4, CPLD (CoolRunner2)

**Embedded Systems:** Bare-Metal Programming, ISR, Timer Programming, Power Management (WFI), Register-Level Programming, UART/USART, GPIO, PWM, ADC, FSM Design, Synchronous Digital Logic, HDL Synthesis, Testbench Simulation

**Edge Computing:** Real-time local processing on embedded Linux (Raspberry Pi 4)

**Robotics & AI:** OpenCV, TensorFlow Litel, Sensor Fusion, Autonomous Navigation

**Software & Tools:** Git, GitHub, GitLab, Linux (Embedded/Raspbian), Agile/Scrum, Unit Testing(Junit), Code Optimisation, Xilinx ISE

**Hardware & Design:** Motor Control (DC/Stepper/Servo), Circuit Design, SolidWorks, Rapid Prototyping

**Currently Learning:** STM32 Bare-Metal Programming (Udemy, 2026), FreeRTOS, C++ (deepening)

## EDUCATION

### MSc Software Engineering with Professional Placement Year — *Sep 2025 - Present*

**Cardiff University** | Cardiff, United Kingdom

Relevant Modules: Web Application Development(Spring Boot, REST APIs, Databases, Security), DevOps (CI/CD, Docker, Terraform, Azure, Jenkins, Virtualisation, Performance Monitoring), Data Manipulation & Analysis (Pandas, NumPy, SQL, Statistical Analysis, Data Pipelines)

### BEng (Honours) Robotics and Mechatronics Engineering — *Oct 2020 - Jan 2025*

**Swinburne University of Technology** | Kuching, Malaysia

Upper Second Class (2:1) | GPA: 3.29/4.00 | Dean's List of Top Achievers (2021)

Relevant Modules: Embedded Microcontrollers, Digital Electronics Design, Circuits & Electronics, Robot System Design, Electrical Actuators & Sensors, Mechatronics Systems Design, Digital Signal Processing

## TECHNICAL PROJECTS

### AI-Powered Autonomous Human-Following Robot — *Sep 2023 - Apr 2024*

**BEng Final Year Project** |

*Stack: Raspberry Pi 4, ESP32, Logitech Webcam, C++ (Arduino), Python, OpenCV, TensorFlow Lite, Linux (Raspbian), UART, PWM, GPIO, SolidWorks, L298N Motor Driver*

- Developed embedded Linux software on Raspberry Pi 4 using an asynchronous detection pipeline, processing real-time camera input locally on-device, eliminating cloud dependency and enabling low-latency marker tracking and motor response
- Trained a custom TensorFlow Lite EfficientDet Lite0 model on 350+ annotated images for real-time marker detection, achieving distance estimation accuracy within 1cm margin using focal-length- based template matching at ranges of 30 -120cm
- Programmed Arduino C++ motor control on ESP32 using PWM-based speed regulation and proportional mapping of marker x-coordinate to differential motor speeds, enabling the robot to track and centre the target within the camera frame

- Implemented UART serial communication between Raspberry Pi 4 and ESP32 to transmit real-time position and distance data, integrating the vision processing pipeline with hardware motor control

**Automatic Water Tank Level Control System** *May 2022 - Jun 2022*
**Embedded Systems Module** |
*Stack: ARM Cortex-M (MK20D), Bare-Metal C, ADC, GPIO, FTM Timer, ISR*

- Developed bare-metal C firmware for automated water level control, engineering an interrupt-driven architecture using the FlexTimer Module (FTM) in Input Capture mode for precise timing and fill rate measurement
- Implemented real-time hardware control by directly manipulating peripheral registers (SIM, PORT, GPIO) enabling low-latency pump actuation without OS overhead.
- Utilised the Wait-For-Interrupt (WFI) low-power instruction during idle monitoring cycles to reduce CPU activity while awaiting the next sensor interrupt.

**Microcontroller-Based Home Security Lighting System** *Apr 2022*
**Embedded Systems Module** |
*Stack: ARM Cortex-M4 (MK20D), Bare-Metal C, ADC, Register-Level Programming*

- Developed bare-metal embedded firmware with ADC-based LUX and temperature sensing, implementing hysteresis control logic that eliminated rapid output toggling and ensured stable system states across varying environmental conditions.
- Configured low-level system registers including Clock Gating and Pin Muxing; utilised Low-Power ADC mode to optimise the balance performance and energy efficiency
- Implemented a field-adjustable configuration interface using GPIO polling, enabling physical push-button tuning of sensor sensitivity and setpoints without firmware re-flash

**VHDL Traffic Light Control System** *Dec 2021*
**Digital Electronics Design Module** |
*Stack: VHDL, Xilinx ISE, CoolRunner2 CPLD, FSM, Synchronous Logic*

- Designed and synthesised a traffic light control system in VHDL targeting CoolRunner2 CPLDs using Xilinx ISE, achieving a clean synthesis with 0 errors and 0 warnings across three interconnected modules: Input Sync Memory, State Machine, and Traffic Top Level.
- Implemented a 10-state, 26-transition Finite State Machine with one-hot encoding, producing Moore-style outputs for traffic light control and Mealy-style outputs for timer management, with state transitions driven by a 4-bit synchronous counter using defined constants for 2 s, 3 s, and 10 s delays.
- Engineered the Input Sync Memory module to clock-synchronise car sensor inputs and implement a latch-based pedestrian button memory, retaining press events until the corresponding state machine clear signal was asserted, ensuring accurate priority handling across EW and NS lanes.
- Developed and executed comprehensive VHDL testbenches covering 6+ scenarios including simultaneous vehicle and pedestrian detection, button-press priority ordering, and input memory retention, verifying correct system behaviour at the module and top-level integration level.

## WORK EXPERIENCE

**STEM Ambassador ( Robotics & Embedded Systems)** *Oct 2025 - Present*
**Cardiff University** | Cardiff, United Kingdom

- Facilitate hands-on robotics workshops for 100+ students, teaching embedded programming concepts including sensor interfacing, actuator control, and control logic using the LEGO SPIKE platform
- Demonstrate practical applications of embedded systems principles, inspiring the next generation of engineers and reinforcing own technical communication skills

**Process Engineer (Industrial Placement)** *Jul 2023 - Aug 2023 | Feb 2025*
**Isuzu East Africa** | Nairobi, Kenya

- Conducted time-motion studies across assembly lines, identifying process bottlenecks and implementing improvements that reduced average operation time by 15%
- Collaborated with cross-functional engineering teams to optimise manufacturing data management systems, improving production traceability and quality control

## CERTIFICATIONS & ACHIEVEMENTS

- Dean's List of Top Achievers - Swinburne University of Technology (2021)
- Dassault Systèmes Additive Manufacturing Certification (2022)
- Currently completing: Embedded Systems Bare-Metal Programming - STM32 (Udemy, 2026)